Automated Detection of Convulsive Seizures Using a Wearable Accelerometer Device

Shitanshu Kusmakar, *Student Member, IEEE*, Chandan K. Karmakar, *Member, IEEE*, Bernard Yan, Terence J.O'Brien, Ramanathan Muthuganapathy, and Marimuthu Palaniswami

***Abstract*—Epileptic seizure detection requires specialized approaches such as video/electroencephalography monitoring. However, these approaches are restricted mainly to hospital setting and requires video/EEG analysis by experts, which makes these approaches resource- and labor-intensive. In contrast, we aim to develop a wireless remote monitoring system based on a single wrist-worn accelerometer device, which is sensitive to multiple types of convulsive seizures and is capable of detecting seizures with short duration. Simple time domain features including a new set of Poincaré plot based features were extracted from the active movement events recorded using a wrist-worn accelerometer device. The best features were then selected using the area under the ROC curve analysis. Kernelized support vector data description was then used to classify nonseizure and seizure events. The proposed algorithm was evaluated on 5576 h of recordings from 79 patients and detected 40 (86.95%) of 46 convulsive seizures (generalized tonic-clonic (GTCS), psychogenic nonepileptic, and complex partial seizures) from 20 patients with a total of 270 false alarms (1.16/24 h). Furthermore, the algorithm showed a comparable performance (sensitivity 95.23% and false alarm rate 0.64/24 h) with respect to existing unimodal and multimodal methods for GTCS detection. The promising results shows the potential to build an ambulatory monitoring convulsive seizure detection system. A wearable accelerometer based seizure detection system would aid in continuous assessment of convulsive seizures in a timely and non-invasive manner.**

***Index Terms*—Accelerometer, generalized tonic-clonic seizures (GTCS), psychogenic non-epileptic seizures (PNES), complex partial seizures (CPS), support vector data description (SVDD), wearable device.**

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S. Kusmakar and M. Palaniswami are with the Department of Electrical and Electronic Engineering, University of Melbourne.

C. K. Karmakar is with the School of Information Technology, Deakin University, Geelong, VIC 3225, Australia (e-mail: karmakar@deakin.edu.au).

B. Yan and T. O'Brien are with the Melbourne Brain Centre Department of Medicine, Royal Melbourne Hospital.

R. Muthuganapathy is with the Department of Engineering Design, Indian Institute of Technology Madras.

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I. Introduction

EPILEPSY is one of the most prevalent neurological disorder with approximately 50 million people living with epilepsy worldwide [1]. Epilepsy is characterized as a chronic neurological disorder leading to recurrent seizures, manifesting as events involving convulsions of a part (*partial*) or complete body (*generalized*). Typically, epilepsy is suspected when there is a history of seizures. A seizure is characterized as epileptic when it is coupled with sudden change in behavior and/or consciousness accompanied by an abnormal electrical discharge in the brain. The gold standard method for detection of epileptic seizures is video-electroencephalography monitoring (VEM). The advantages offered by VEM get limited owing to its availability only in tertiary and super-specialty hospitals. Thus, leading to an impaired quality of life (QOL) for patients with low-socio economic status. Epileptic seizures can be categorized into two types: convulsive and non-convulsive. Seizures with an associated motor component are termed as convulsive seizures [2]. Convulsive seizures have shown to be the major contributor to seizure associated accidents, injuries, and sudden unexpected death in epilepsy (SUDEP) [3]. Continuous non-invasive assessment of convulsive seizures using a wearable device will aid in improving the QOL, and reducing the risk of injury and fatal harm to patients with epilepsy.

Several wearable seizure detection systems based on measuring different physiological parameters have been proposed [4]. The most commonly monitored physiological parameters are: electroencephalography (EEG), electrocardiography (ECG), electrodermal activity (EDA), and surface electromyography (sEMG). Although, EEG is the gold standard method for detection of seizures, EEG based ambulatory monitoring systems requires the patient to wear scalp electrodes at all times. Thus, the system leads to stigmatization and discomfort to patient, rendering such systems impractical in ambulatory setting [5]. Few studies have also used ECG derived heart rate variability (HRV) to detect generalized tonic-clonic seizures [6]. However, HRV is a low specificity signal and the heart rate changes vary from person to person [7]. Thus, ECG based seizure detection systems are patient specific and require model parameters to be tuned for individual patient. Recently, autonomic changes in sympathetically mediated electrodermal activity (EDA) were also shown to correlate with seizures [8]. Approaches utilizing EDA have shown promising results, however due to the latency in onset of EDA these systems perform better only when

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coupled with accelerometers [9]. In addition, EDA measurements can be uncomfortable or obtrusive due to the application of direct current required to measure EDA. Researchers have also investigated the use of sEMG in automated detection of seizures. sEMG signal shows a high specificity as a seizure marker [10]–[12]. However, sEMG based systems require continuous use of surface electrodes which might lead to discomfort, in addition to being susceptible to movement artifacts [13].

A more comfortable and non-obtrusive seizure detection system for daily living would be an electrode-less system. Convulsive seizures have an associated motor component and accelerometer sensors placed on the limbs can detect these movements with minimal intrusion to the patient [4]. Several studies have shown the use of accelerometer sensors in detection of seizures [9], [10], [14]–[18]. The performance of these seizure detection systems varies across studies with respect to the type of seizure, number and type of sensors, and the minimum duration of seizure that can be detected. There is an inherent trade-off between minimum seizure duration and false alam rate (FAR) of a seizure detection system [10]. As seizure duration is increased, the FAR decreases and sensitivity increases. This factual increase in the performance of the system is at the cost of missing seizures that are shorter than the minimum specified duration.

Although convulsive seizures with shorter duration may not pose the SUDEP associated threat as posed by seizures with duration >1 min, the chances of injury and harm still remains high [3]. Thus, it is important to detect convulsive seizures of shorter duration. In this study, we have selected a minimum seizure duration of 10 s based on the aforementioned non-EEG based seizure detection studies. However, all the studies with a minimum seizure duration of 10 s either employ: (1) multiple sensors [10], or (2) a multi-modality system [9]. These systems have inherent drawbacks that were discussed in the previous paragraphs. Therefore, in this study we propose to use a single wrist-worn accelerometer device for detecting seizures of duration as short as 10 s.

Furthermore, research in this space tends to focus only on the detection of generalized tonic-clonic seizures (GTCS) [19], though there is no clear consensus that patients having GTCS events are at a higher risk or are directly related to epilepsy associated mortality [20]. A high number of patients admitted to our epilepsy monitoring unit were found to have convulsive psychogenic non-epileptic seizures (PNES) [21]. The burden of PNES on healthcare is equivalent to GTCS [21]. In our study cohort, we had patients with another uncommon class of seizures that are often missed as behavioral changes and at times manifest as repetitive movement of upper limbs [22]. These seizures are referred as complex partial seizures (CPS). In addition, few patients also had comorbid condition i.e., having PNES events in addition to a confirmed diagnosis of epilepsy. This indicates that the seizure detection systems should be able to detect other seizure types in addition to GTCS. Thus, in this study, we aim to propose a method that can detect different types of convulsive seizures from single wrist-worn accelerometer sensor.

The continuous monitoring devices such as wearable accelerometer sensors collect an unprecedented amount of movement data. Seizures are rare events thus, the accelerometer data is dominated by non-seizure events which makes the majority class. Since the proposed approach is targeting detection of seizures with duration ≥10 s, it inherently increases the degree of class imbalance in the data. Various algorithms ranging from threshold or rule based heuristics [17], [23], [24] to machine learning based approaches [10], [25], [26], have been proposed for automated detection of seizures using accelerometer based devices. The class imbalance has rarely been mentioned in these studies however, some studies have explicitly outlined this characteristic of the accelerometry data [9], [14]. Cuppens *et al.* [14], presented an approach based on estimating the probability density function (PDF) of non-seizure events. A 95% threshold on PDF was then used to classify events as outliers or seizures. However, their approach was patient specific as the PDF and the thresholds had to be estimated for every patient. In contrast, Poh *et al.* [9] proposed a non-patient specific system based on SVM. Both [9] and [14] have shown fairly high sensitivity (94% and 95.24%, respectively) in detecting seizures. However, [9] uses SVMs for classification that may lead to suboptimal results on imbalanced data set, and the classification performance decreases with the degree of imbalance in the data [27]. Whereas, in [14] they have not reported the FAR, which makes it difficult to assess the seizure detection performance of their model on imbalanced datasets. Thus, it may be difficult to generalize these models on other seizure datasets. Therefore, a technique that can combat the class imbalance would be more suited for seizure detection using accelerometry. In this work, we present an automated seizure detection approach based on kernelized support vector data description (SVDD) that can handle the class imbalance in the data [28].

In our previous work [29], we had presented a system for detection and classification of PNES from GTCS. However, the main aim of that work was distinguishing PNES from GTCS rather than optimizing the seizure detection system. Whereas in this study, we aim at improving the detection of seizure events from non-seizure events. Characterizing seizure and non-seizure events is very critical in improving the device efficacy, since it directly controls the sensitivity and FAR of the system. Besides, in this work we have included convulsive CPS in the seizure event cohort. In this study, we present the performance evaluation of a novel seizure detection algorithm that is sensitive to different seizure types (GTCS, convulsive PNES and CPS) and is capable of detecting seizures with duration ≥10 s. Moreover, the proposed approach is developed using the data from a single wrist-worn accelerometer device, which makes it befitting to ambulatory monitoring of epileptic patients. The major contributions of the this work are:

- In addition to GTCS events, we also looked at convulsive PNES and CPS events. These are two uncommon class of seizures that have rarely found their mention in accelerometer based seizure detection system.
- The proposed system is sensitive to convulsive seizures with duration as short as 10 s.

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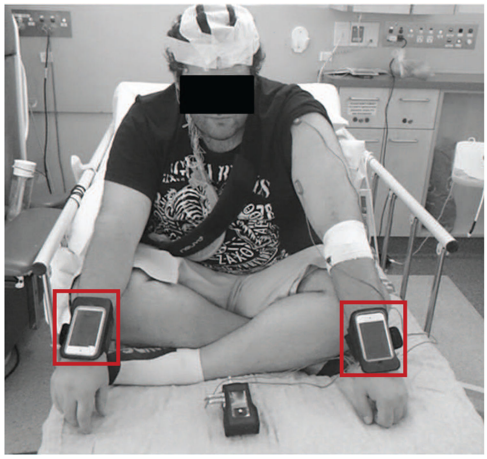

Fig. 1. A participant at the long-term epilepsy monitoring unit at Royal Melbourne Hospital. The wireless accelerometer devices are indicated by the rectangles. The data from only one device was used to develop the seizure detection algorithm.

- The proposed system is based on a single wrist-worn accelerometer device. Thus, eliminating any discomfort or stigmatization to the patient as a result of multiple sensors and electrodes.
- A thorough comparison of the proposed approach with the existing studies.

II. Method

A. Data Acquisition

Apple iPod Touch devices with inbuilt microelectromechanical systems (MEMS) accelerometer sensor were used to collect the motor data from patients undergoing long-term VEM at Royal Melbourne Hospital. Each device consists of a 3D accelerometer with a sensitivity of ±2.5 g. A smart application was developed to continuously record 3D accelerometry data with a time stamp. The wireless accelerometer devices were firmly strapped on both wrists such that the devices were in contact with the dorsal side of forearms (Fig. 1). The accelerometer recordings were sampled at 50 Hz and synchronized with VEM by manually updating the time from same network. The accelerometer data was recorded continuously and stored on the flash memory of the devices which was later extracted for $off-line$ analysis. Accelerometer data was collected for the complete duration of VEM. Each recording session lasted for 12 hours after which the devices were replaced with similar charged devices. During the monitoring period, patients were not constrained to remaining in the bed and performed different activities of daily living (ADL) while wearing the accelerometer devices. The study was approved by human research ethics committee of the Royal Melbourne Hospital (HREC Project 300:259).

B. Patients and Types of Seizure Events

A total of 79 patients were recruited in the study spanned over 2012 to 2015. Out of the 79 patients, 20 (25.31%) had convulsive seizures, 15 (18.98%) had non-convulsive seizures while, 44 patients did not have any seizure during the monitoring period. Based on a retrospective analysis each seizure event was reviewed by expert epileptologists for it's type, onset, offset, seizure semiology as observed on VEM, and localization while being blinded to accelerometer traces. Out of the 20 patients with convulsive seizures, 10 (50%) only had GTCS events, 6 (30%) only had PNES events, 2 (10%) only had CPS events while, 1 patient each had GTCS + PNES, and GTCS + CPS, respectively. This study focuses only on detection of convulsive seizures and each convulsive event was annotated using custom software written in MATLAB 2015b (MathWorks, Natick, MA, U.S.A.). In this study, we have included the accelerometer data from only one wrist-worn device as wearing two devices may be inconvenient for the patient during continuous ambulatory monitoring. The left hand data was the default choice unless the data was unavailable or corrupted. Out of total 79 patients, right hand data from only 4 patients was used. In addition, an analysis was also performed by considering the right hand data of patients. The detailed results of the right hand data are presented in the supplementary material.

TABLE I
Overview of Patients With Convulsive Seizures

Patient	Age	#GTCS	#PNES	#CPS	Record duration
P_1	28	1	0	0	86 : 17 : 53
P_2	35	1	0	0	93 : 36 : 17
P_3	20	1	0	0	64 : 32 : 54
P_4	19	6	0	0	86 : 45 : 08
P_5	30	2	1	0	81 : 42 : 41
P_6	57	2	0	0	90 : 15 : 35
P_7	42	2	0	0	88 : 33 : 43
P_8	51	1	0	0	86 : 00 : 19
P_9	32	1	0	0	87 : 17 : 21
P_{10}	22	1	0	0	64 : 53 : 48
P_{11}	20	2	0	1	85 : 38 : 30
P_{12}	19	1	0	0	50 : 28 : 09
P_{13}	36	0	0	3	20 : 28 : 50
P_{14}	22	0	0	1	48 : 00 : 20
P_{15}	59	0	5	0	60 : 38 : 44
P_{16}	22	0	1	0	70 : 13 : 28
P_{17}	19	0	9	0	45 : 29 : 43
P_{18}	37	0	1	0	50 : 40 : 38
P_{19}	43	0	1	0	62 : 20 : 49
P_{20}	20	0	2	0	73 : 51 : 20
#20 Patients	31.65 ± 12.73	21	20	5	1397 : 46 : 40

Record duration format hh:mm:ss

The patient cohort included patients with different types of convulsive seizures - GTCS, PNES, and CPS (Table I). GTCS events are characterized by a hyper-synchronous electrical activity that spreads to the entire brain. GTCS events involve involuntary muscle spasm (*tonic phase*) generally with a loss of consciousness, followed by unprovoked convulsions of the limbs (*clonic phase*). PNES events are paroxysmal events that visually resemble GTCS events however, PNES are not caused by epileptiform discharge in brain. On the other hand, CPS events are characterized as focal seizures causing partial impairment of consciousness that at times is accompanied by unusual automatisms.

The total recording duration for 79 patients is 5576 h. However, 20 patients who had atleast one seizure, correspond to

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TABLE II
TABLE SHOWS THE NUMBER OF OBSERVED SEIZURES AND THE NUMBER OF PATIENTS CORRESPONDING TO DIFFERENT MINIMUM SEIZURE DURATION

Seizure duration	$\geq 10s$	$\geq 20s$	$\geq 30s$	$\geq 40s$	$\geq 50s$	$\geq 60s$
# Patients	20/79	20/79	18/79	18/79	17/79	14/79
# Observed seizures	46/46	45/46	39/46	34/46	29/46	25/46

A total of 20 out of 79 patients had 46 convulsive seizures during the monitoring period.

approximately 1397 h of recording. A total of 46 convulsive seizures (21 GTCS, 20 PNES, and 5 CPS) were observed from 20 patients who had seizures (Table I). The distribution of the observed seizures corresponding to the minimum seizure duration is shown in Table II. The number of observed seizures decreases as minimum seizure duration increases (Table II). The dataset is divided into four groups mainly: 1) Patients with only GTCS ($PG1$), 2) Patients with GTCS and CPS ($PG2$), 3) Patients with GTCS and PNES ($PG3$), 4) Patients with GTCS, PNES and CPS ($PG4$). In addition, the CPS and PNES patients were also analyzed individually in groups of patients: 1) with only CPS events (PG_{CP}), and 2) with only PNES events (PG_{NE}) (refer to Table I in the supplementary material). This allows us to understand the motor behavior of different seizure types by comparing the performance of the algorithm across groups.

C. Pre-Processing

The movement data recorded using wrist-worn accelerometer device is dominated by non-seizure and no movement data. Thus, pre-processing steps which reduces the volume of the data, and the computational workload, are vital for detection systems based on supervised learning. This pre-processing is described in our previous work [29] and will be explained herewith.

In the pre-processing step, the accelerometer data is analyzed in windows of 10 s with 50% overlap. These two values, length and overlap of windows were chosen as the minimum seizure duration considered in this work is 10 s. The pre-processing involves two steps namely, activity and time filtering. The activity filter selectively filters out all epoch's with no or subtle activity. To explain the activity filter, let us represent the accelerometer data along one Cartesian co-ordinate by a discrete time signal $\{x[n]\} = \{x_1, x_2, x_3, \dots\}\ \forall\ n \in \mathbb{Z}$. At, any time interval t, the accelerometer signal can then be represented as $x[t]|_{t=n\Delta t}$, where Δt is the sampling period of the accelerometer signal.

In every time window, a threshold of 0.2 g is then applied on the accelerometer signal to discard samples below the specified threshold as no movement or no activity data. The threshold is chosen empirically based on the lower bound of the collected seizure data. The thresholding step can be explained as follows:

$$x'[t] = \begin{cases} x[t] & |x[t]|_{t=n\Delta t}| \geq 0.2g \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where, $x'[t]$ is the signal obtained after thresholding.

After the thresholding step, the accelerometer signal is again filtered to remove any artifacts or spurious spikes using a sixth order Butterworth bandpass filter with cutoff frequency in range 2 25 Hz. The cutoff frequency is chosen based on the analysis of dominant frequency during typical seizure events as seen from our previous works [2], [29]. Followed by the filtering step, all windows with more than 50% activity are labeled as epochs with sustained activity or active epochs. The activity of the signal in a window is calculated as follows:

$$A_n = \frac{1}{(10 \times F_s)} \sum_{t=l_1}^{l_2} |\text{sgn}(x'[t])| \tag{2}$$

where, A_n represent the activity value in the nth window, F_s is the sampling frequency of the accelerometer signal, $l_1 = (n-1) \times \frac{W_{len}}{2} \times F_s + 1$, $l_2 = (n+1) \times \frac{W_{len}}{2} \times F_s$, $W_{len} = 10$, and sgn is the signum function that can be explained as:

$$\text{sgn}(x'[t]) = \begin{cases} 1 & x'[t] \geq 0 \\ -1 & x'[t] < 0 \end{cases} \tag{3}$$

After the activity filtering step, all consecutive 10 s epochs that are labeled as active epochs and are lying <20 s apart are merged as a single "event". This merging of consecutive epochs for every patient leads to several events of varied duration that are termed here as active events. These events includes both: (1) non-seizure events/ADL, and (2) seizure events. This step is followed by a time filtering step that removes all events with duration lesser than the minimum seizure duration. For every patient, the pre-processing steps segment the whole data into several events that have a duration $\geq$10 s. After pre-processing the data corresponding to all 79 patients, we obtain a total of 6,814 events that includes: (1) 6,768 non-seizure events, and (2) 46 seizure events. For every patient, features are extracted from all events and each event is classified into a non-seizure or seizure event using a binary state classifier.

D. Feature Extraction and Selection

Various features are extracted from the movement events obtained after the pre-processing step. Features are extracted from four accelerometry channels (a_x, a_y, a_z and, a_{norm}) where, $a_{\text{norm}} = \sqrt{a_x^2 + a_y^2 + a_z^2}$. Time domain features that are commonly used in the detection of seizures using accelerometry [29], [14], are then calculated for classification of seizure and non-seizure events. Time domain features have shown to be the best features for detection of seizures using accelerometry [30]. Moreover, time domain features offer the advantage of low computational complexity which is important in context of real-time analysis [31]. Since we are considering three different types of convulsive seizures we need a robust set of features to classify seizure and non-seizure events. Therefore, in addition to accelerometry features used commonly in the detection of convulsive seizures we used a new set of computationally efficient time domain features, extracted using Poincaré analysis. The Poincaré based features give non-linear and dynamical information, and have been extensively used in biomedical applications [32]. Poincaré plot is a geometrical representation of the time series data when, plotted consecutively in the Cartesian plane. This technique allows the accelerometer signal to be represented in the state (phase) space or Cartesian plane by

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TABLE III
LIST OF FEATURES CALCULATED ACROSS EACH CHANNEL ($a_x, a_y, a_z, a_{\text{norm}}$), WHERE '×' REPRESENTS THE SELECTED FEATURES

Features	Equations	a_x	a_y	a_z	a_{norm}
Inter quartile range	$(Q_3 - Q_1)$				×
Standard deviation	$\frac{1}{N-1}\sum_{t=1}^{N}\left(a[t]-\bar{a}\right)^2$	×	×	×	×
Maximum amplitude	$\max_t\left(\lvert a[t]\rvert\right)$				×
Mean	$\frac{1}{N}\sum_{t=1}^{N} a[t]$				×
Median	$median(\lvert\{a[t]\}\rvert)$				
Zero crossing rate	$\frac{1}{2N}\sum_{t=1}^{N}\left\lvert sgn(a[t]) - sgn(a[t-1])\right\rvert w(n-t)$	×	×	×	
Kurtosis	$\frac{E(a-\mu)^4}{\sigma^4}$				
Skewness	$\frac{E(a-\mu)^3}{\sigma^3}$				
Total average power	$\frac{1}{N}\sum_{t=1}^{N}(\lvert a[t]\rvert^2)$	×	×	×	×
SD1†	$SD1^2 = \frac{1}{2}SDSD^2$	×	×	×	×
SD2†	$SD_2^2 = 2SDA^2 - \frac{1}{2}SDSD^2$				
Ratio†	$\frac{SD_1}{SD_2}$				×
CCM†	$CCM_m = \frac{1}{C}\sum_{i=1}^{N-2}\lvert A(i)\rvert$		×		×

where $\{a[t]\}$ is the accelerometer signal, Q_1 and Q_3 are the first and the third quartile; sgn is the signum and w(n) is the windowing function; *SDSD* is the standard deviation of the successive differences at lag $m = 1$; $SDSD = \sqrt{E[(a_m - a_{m+1})^2]}$; *SDA* is the standard deviation at lag m $SDA = \sqrt{E[a_m^2] - (E[a_m])^2}$; $A(i)$ area of triangle and C is a constant. $|\cdot|$ indicates absolute value.
† Poincaré descriptors.

constructing two dimensional plots of the time series data. Descriptors such as SD1, SD2 and CCM are calculated from the Poincaré plot at lag $m = 1$ [33]. The best features are then selected by performing a receiver operating characteristic (ROC) analysis [34] where, the area under the curve (AUC) of an individual feature was determined to asses the class separation. As the number of events and the types of seizures vary across the patients (Table I), an ensemble approach was utilized for feature selection. In this approach, the feature selection algorithm was repeated 20 times in a leave-one-patient-out fashion. In each iteration the importance of each feature was evaluated using training data (data from 19 of 20 patients) by calculating AUC value to distinguish seizure from non-seizure events. Features having an AUC value ≥ 0.90, and that appeared for at least 15 (75%) of total 20 iterations were selected. Table III shows the list of extracted features, where, '×' represents the selected features (21 features) for event detection across each accelerometer channel.

E. Support Vector Data Description

In this study, we have used a state-of-art classifier called support vector data description (SVDD) inspired from the support vector machine (SVM) family. SVDD builds a description on the non-seizure events and this boundary is used to detect any novel or uncharacteristic events, which represent seizures. SVDD obtains a hyper-spherical decision boundary, that separates the non-seizure and seizure events. By controlling the volume of this sphere we can control the sensitivity and FAR of the detection system. The optimization problem, can be formulated as the minimization of the volume of a sphere with center a and radius $R > 0$. Let the training set be represented by $\{x_i\}$ where, $\{x_i\}$ is a d-dimensional feature vector $\forall\ 1 \leq i \leq N$, and N denotes the number of events. To have a robust seizure detection system, we tighten the description by minimizing the volume of sphere to prevent any chances of accepting an outlier or a seizure event as a non-seizure event. The error function to be minimized is represented by (4).

$$\min_{R,a,\xi_i} \quad F(R,a) = R^2 + C\sum_i \xi_i \tag{4}$$

With constraints as shown in (5) so that, all events are within the sphere;

$$||x_i - a||^2 \leq R^2 + \xi_i, \quad \xi_i \geq 0 \tag{5}$$

The slack term ξ_i relaxes the penalty on the ith training event to be quantified as an outlier, while cost C controls the trade-off between the hypersphere volume and errors for the training events. The minimization problem (4) can be solved by introducing the Lagrange multipliers α_i and maximizing with respect to α_i to get:

$$L = \sum_i \alpha_i(x_i.x_i) - \sum_{i,j}\alpha_i\alpha_j(x_i.x_j), \tag{6}$$

$$\text{s.t.} \quad \sum_i \alpha_i = 1 \quad a = \sum_i \alpha_i x_i \quad 0 \leq \alpha_i \leq C \tag{7}$$

α_i takes the value 0 if the event x_i is within the sphere. If the event x_i lie on the decision boundary α_i takes the value between 0 and C and these events are called the support vectors (SV's) of the description. On the other hand, if an event x_i lies outside the hypersphere then the events are classified as outliers or in our case as seizure events with α_i taking the value C. This describes the categories in which an event can be classified according to Kuhn-Tucker conditions.

Now, to test an event y its distance from the center of the sphere has to be determined and the event is accepted if it falls within the sphere, as shown by (8).

$$||y - a||^2 = (y.y) - 2\sum_i \alpha_i(y.x_i) + \sum_{i,j}\alpha_i\alpha_j(x_i.x_j) \leq R^2 \tag{8}$$

Further, the decision boundary can be made flexible by the use of Kernel functions. This can be generalized by replacing the inner products (x, z) by a kernel function $K(x, z) = (\phi(x)\phi(z))$, where K represents a Kernel function which maps the input data to a higher dimension $\phi(\cdot) : x^d \rightarrow x^{d_h}$ where, $(d_h \geq d)$. Thus,

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(6), (7), and (8) will now be represented as:

$$L = \sum_i K(x_i.x_i) - \sum_{i,j} \alpha_i \alpha_j K(x_i.x_j) \tag{9}$$

$$\text{s.t.} \quad \sum_i \alpha_i = 1 \quad a = \sum_i \alpha_i \phi(x_i) \quad 0 \le \alpha_i \le C \tag{10}$$

$$||y - a||^2 = K(y.y) - 2\sum_i \alpha_i K(y.x_i) + \sum_{i,j} \alpha_i \alpha_j K(x_i.x_j) \le R^2 \tag{11}$$

In our work, we have chosen a radially symmetrical Gaussian kernel function as it has been shown to results in good description of the data using SVDD [35]. The radially symmetric Gaussian kernel function is represented by (12), which can be completely specified using a single parameter γ.

$$K(x_i.x_j) = \exp(-\gamma||x_i - x_j||^2) \quad \gamma > 0 \tag{12}$$

F. Model Estimation and Validation

In this study, we have reported patient specific performance as well as overall performance. For evaluating patient specific performance, we trained the algorithm on data from patients who had at least one seizure during the monitoring period. Let P_{SZ} denote the set of patients with seizures, and let P_{WS} denote the set of patients without any seizure events. Then the data set used for learning the classification model is represented by patient set P_{SZ}. To develop a generic non-patient specific algorithm, we employed a leave-one-patient-out cross validation approach that excludes all data of the test patient from the training phase. Let P_{LO} denote the left-out patient. Then the training set ($\text{Train}_{\text{set}}$) is represented by patient set $P_{SZ} - P_{LO}$. The training set is further divided into a learning ($\text{Train}_{\text{learn}}$) and a validation set ($Val_{\text{set}}$). Two-thirds of the training data is randomly selected as the learning set and one-third as validation set with approximately equal class proportion. This process is repeated in ten randomizations to tune the SVDD parameters. Fig. 2 shows the output of the training phase for one randomization, on patient group with different types of seizures.

There are two parameters to be tuned for SVDD, C and the kernel parameter γ. C is a trade-off between errors and model complexity. A large value of C results in a tighter decision boundary whereas, a smaller value of C relaxes the decision boundary. A tighter decision boundary leads to an increased seizure detection sensitivity but, at the cost of higher FAR. On the other hand γ controls the smoothness of the decision boundary. A smaller value of γ will not be able to capture the overall complexity or the shape of the data, whereas a large value can lead to overfitting. Thus, for an optimal solution the parameters C and γ have to be tuned to minimize the trade-off between detection sensitivity and FAR. As in [14], we have wrapped a custom cost function (13) with the classifier, which applies a higher penalty for misclassifying a seizure event.

$$\text{cost}(C, \gamma) = -w \times \text{sensitivity}(C, \gamma) - ppv(C, \gamma) \quad w = 2 \tag{13}$$

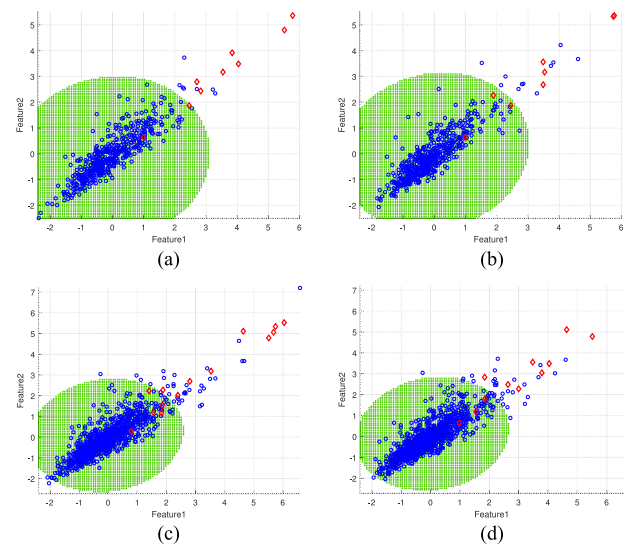

Fig. 2. Output of the training phase shown for one randomization of the training phase. The region in green shows the hyper-spherical decision boundaries obtained using support vector data description (SVDD). For visualization purposes, only two features (mean and standard deviation of the resultant signal) are used. The non-seizure events are shown by ○ (blue), and seizure events are shown by ◇ (red). nSV represents the number of support vectors. (a) Only GTCS (nSV = 71). (b) GTCS and CPS (nSV = 95). (c) GTCS and PNES (nSV = 224). (d) GTCS, PNES, and CPS (nSV = 232).

Ten models were obtained from the ten randomizations during validation phase. The model with the least cost is then selected as the optimum model, to evaluate the performance on the test patient P_{LO}. The performance is characterized in terms of sensitivity, specificity, $F_1 score$ metric, positive predictive value (PPV), and the FAR (false alarms per 24 h) of the system.

To evaluate the generalization of the proposed algorithm in terms of $FAR/24$ h, we tested the algorithm on an independent population. We utilized the data from the patients who did not have any seizures during the monitoring period (patient set P_{WS}). For this, a SVDD classifier is trained using the data from patient set P_{SZ}. The SVDD parameter tunning is performed by randomly selecting two-third of the data as learning set and one-third as validation set. The procedure is repeated in ten randomizations and the model with the least cost (equation (13)) is selected for performance evaluation on data from patient set P_{WS}. The details of the performance metrics and histogram plots of $FAR/24$ h for patients without any seizures (P_{WS}) are shown in the supplementary materials. In addition, the pseudo-codes explaining the training and testing procedure, are also presented in the supplementary materials.

III. Results

The leave-one-patient-out validation results for different patient groups are shown in Table IV(a)–IV(d). The results of patient group $PG1$ (Table IV(a)) shows that the SVDD based GTCS detection system could correctly identify 20 out of the 21 GTCS events (mean sensitivity 95.83%, 95% confidence interval [CI] 87.50–100%) and 1504 (98.04%) out of 1534 non-seizure events with a mean FAR of 0.72/24 h (30 false alarms

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TABLE IV
THE LEAVE-ONE-PATIENT-OUT (LOPO) VALIDATION RESULTS SHOWN FOR THE DIFFERENT PATIENT GROUPS

Patient ID	Sens. (%)	Spec. (%)	F_1score (%)	PPV (%)	$FAR/24h$
P_1	100.00	92.11	40.00	25.00	0.83
P_2	100.00	98.05	28.57	16.67	1.28
P_3	100.00	98.97	50.00	33.33	0.70
P_4	100.00	95.71	80.00	66.67	0.82
$P_5^{¶}$	100.00	92.94	40.00	25.00	1.76
P_6	100.00	100.00	100.00	100.00	0.00
P_7	100.00	98.51	80.00	66.67	0.27
P_8	100.00	96.34	40.00	25.00	0.83
P_9	100.00	99.47	50.00	33.33	0.50
P_{10}	100.00	98.81	66.67	50.00	0.36
$P_{11}^{¶}$	50.00	96.25	33.33	25.00	0.84
P_{12}	100.00	98.98	66.67	50.00	0.47
Overall	95.23 (20/21)	98.04 (1504/1534)	56.33	40.00	0.72

¶ P_5 and P_{11} had multiple types of seizures (Table I). Only seizure events that are specific to the group are considered as a seizure during training, while other events are considered as non-seizure events.
¶ Undetected event: 1 GTCS event for patient P_{11}.

(a)

Patient ID	Sens. (%)	Spec. (%)	F_1score (%)	PPV (%)	$FAR/24h$
P_1	100.00	92.11	40.00	25.00	0.83
P_2	100.00	98.05	28.57	16.67	1.28
P_3	100.00	97.95	33.33	20.00	1.48
P_4	100.00	95.71	80.00	66.67	0.82
P_5	100.00	91.76	36.36	22.22	2.00
P_6	100.00	97.09	57.14	40.00	0.79
P_7	100.00	91.04	40.00	25.00	1.62
P_8	100.00	95.12	33.33	20.00	1.11
P_9	100.00	99.47	50.00	33.33	0.50
P_{10}	100.00	98.81	66.67	50.00	0.36
$P_{11}^{†}$	66.67	89.87	30.77	20.00	2.24
P_{12}	100.00	98.98	66.67	50.00	0.47
$P_{13}^{†}$	66.67	93.57	57.14	50.00	2.34
$P_{14}^{†}$	0.00	91.89	0.00	0.00	1.5
Overall	88.46 (23/26)	96.75 (1550/1602)	45.54	30.66	1.24

† Undetected events: 1 CPS event for patient P_{11}, 1 CPS event for patient P_{13}, and 1 CPS event for patient P_{14}.

(b)

Patient ID	Sens. (%)	Spec. (%)	F_1score (%)	PPV (%)	$FAR/24h$
P_1	100.00	92.11	40.00	25.00	0.83
P_2	100.00	98.05	28.57	16.67	1.28
P_3	100.00	95.90	20.00	11.11	1.48
P_4	100.00	95.71	80.00	66.67	0.82
P_5	100.00	92.86	50.00	33.33	1.76
P_6	100.00	97.09	57.14	40.00	0.79
P_7	100.00	98.51	80.00	66.67	0.27
P_8	100.00	92.68	25.00	14.29	1.67
P_9	100.00	99.47	50.00	33.33	0.50
P_{10}	100.00	96.43	40.00	25.00	1.1
P_{11}	100.00	91.25	36.36	22.22	1.96
P_{12}	100.00	97.96	50.00	33.33	0.95
P_{15}	100.00	94.06	62.50	45.45	2.37
$P_{16}^{†}$	0.00	94.87	0.00	0.00	1.36
$P_{17}^{†}$	66.67	98.00	70.59	75.00	1.05
P_{18}	100.00	98.74	40.00	25.00	1.42
$P_{19}^{†}$	0.00	99.08	0.00	0.00	1.53
P_{20}	100.00	96.38	19.05	10.53	5.52
Overall	87.80 (36/41)	97.12 (2872/2957)	44.44	29.75	1.56

† Undetected events: 1 PNES event for patient P_{16}, 3 PNES events for patient P_{17}, and 1 PNES event for patient P_{19}.

(c)

Patient ID	Sens. (%)	Spec. (%)	F_1score (%)	PPV (%)	$FAR/24h$
P_1	100.00	89.47	33.33	20.00	1.11
P_2	100.00	97.28	22.22	12.50	2.82
P_3	100.00	97.44	28.57	16.67	1.85
P_4	100.00	95.71	80.00	66.67	0.82
P_5	100.00	88.10	37.50	23.08	2.93
P_6	100.00	96.12	50.00	33.33	1.06
P_7	100.00	95.52	57.14	40.00	0.81
P_8	100.00	92.68	25.00	14.29	1.67
P_9	100.00	99.47	50.00	33.33	0.54
P_{10}	100.00	91.67	22.22	12.50	2.58
$P_{11}^{†}$	66.67	89.87	30.77	20.00	2.24
P_{12}	100.00	96.94	40.00	25.00	1.42
P_{13}	100.00	84.38	54.55	37.50	5.85
P_{14}	100.00	75.68	18.18	10.00	4.49
P_{15}	100.00	94.06	62.50	45.45	2.37
$P_{16}^{†}$	0.00	94.87	0.00	0.00	1.36
$P_{17}^{†}$	66.67	97.00	66.67	66.67	1.58
P_{18}	100.00	98.74	40.00	25.00	1.42
$P_{19}^{†}$	0.00	98.62	0.00	0.00	2.3
P_{20}	100.00	96.11	18.18	10.00	5.84
Overall	86.95 (40/46)	96.16 (2909/3025)	39.60	25.64	2.25

† Undetected events: P_{11} 1 CPS, P_{16} 1 PNES, P_{17} 3 PNES, and P_{19} 1 PNES.

(d)

from 12 patients). The overall PPV of 40% and a f-score of 56% further shows the efficacy of the seizure detection system. The validation on 67 patients (without any GTCS events) led to 119 false alarms, resulting in an overall FAR of 0.64/24 h (149 false alarms in 5,576 h of recording from 79 patients).

Table IV(c) shows the validation results for patient group $PG3$. The SVDD based approach could detect 36 out of 41 seizure events (mean sensitivity 87.03%, 95% [CI] 77.43–97.04%) and 2872 (97.12%) of 2957 non-seizure events with a mean FAR of 1.56/24 h (85 false alarms from 18 patients). A total of five events (all PNES) were missed by the algorithm corresponding to patients P_{16}, P_{17}, and P_{19}. The validation on 61 patients (without any GTCS or convulsive PNES events) led to 159 false alarms, resulting in an overall FAR of 1.05/24 h (244 false alarms in 5,576 h of recording from 79 patients).

The validation results for the patient group $PG2$ (Table IV(b)) shows that a total of 23 out of 26 seizure events were detected (mean sensitivity 88.09%, 95% [CI] 78.66–98.18%) and 1550 (96.75%) of 1602 non-seizure events were classified correctly. An overall f-score of 45.45% and PPV of 30.66% was achieved with a mean FAR of 1.24/24 h (52 false alarms from 14 patients). Three events were missed corresponding to patients P_{11}, P_{13}, and P_{14}. The validation on 65 patients (without any GTCS or convulsive CPS events) led to 127 false alarms, resulting in an overall FAR of 0.77/24 h (179 false alarms in 5,576 h of recording from 79 patients).

Table IV(d) shows the validation results for patient group $PG4$, having all the three seizure types (GTCS, PNES and CPS). For patient group $PG4$, the proposed algorithm could detect 40 out of 46 seizures (mean sensitivity 86.66%, 95% [CI] 74.23–96.95%) with a mean FAR of 2.25/24 h (116 false alarms from 20 patients). The algorithm also classified correctly 2909 (96.16%) of 3025 non-seizure events. A total of six events (1 CPS and 5 PNES) were missed by the algorithm corresponding to patients P_{11}, P_{16}, P_{17} and P_{19}. The validation on 59 patients (without convulsive seizures) led to 154 false alarms, resulting

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in an overall FAR of 1.16/24 h (270 false alarms in 5,576 h of recording from 79 patients).

Furthermore, validation results corresponding to the right hand data are also presented (see Table III(b) in supplementary material). In line with the aim of the study, the analysis for right hand data is only performed corresponding to patient group $PG4$. For only 19 of total 79 patients, seizures were captured on the right hand device. A total of 51 convulsive events were recorded from the right hand (see Table II in supplementary material). The proposed algorithm detected 43 (84.31%) of 51 convulsive seizures with a mean FAR of 2.90/24 h (138 false alarms from 19 patients). The missed events corresponded to patients P_{16}, P_{17}, P_{18}, and P_{20}. The validation on 60 patients (without convulsive seizures) led to 171 false alarms, resulting in an overall FAR of 1.33/24 h (309 false alarms in 5,563 h of recording from 79 patients).

It is evident from the results shown in Table IV(a)–IV(d), that the best performance (sensitivity, FAR) was obtained for patient group $PG1$. The overall performance starts to drop as patients with other seizure types were included in the group. The sensitivity decreases and FAR increases as we move from patient group with only GTCS ((mean 95.83%, 95% [CI] 87.50–100%), mean FAR (0.72/24 h)) to patient group with all three seizure types ((mean 86.66%, 95% [CI] 74.23–96.95%), mean FAR (2.25/24 h)).

IV. Discussions

In this study, we investigated the use of single wrist-worn accelerometer sensor for detection of convulsive epileptic seizures. The data set analyzed in this work comprised of three types of convulsive seizures: 1) generalized tonic-clonic seizure, 2) psychogenic non-epileptic seizures, and 3) complex partial seizures that manifested as clonic stereotype. Although co-existence of multiple seizure types in a single patient are rare (1 patient with GTCS + PNES and 1 patient with GTCS + CPS (Table I)), existence of PNES and CPS alone are not that rare in individual patients (6 out of 20 patients have PNES and 2 out of 20 have CPS (Table I)). Thus, it is important to detect all of them for analyzing general epileptic cohort for detecting seizures. Since patients with convulsive seizures have a high risk of injury during a seizure event, an effective seizure detection system would be able to raise an alarm on detecting any type convulsive seizure. The seizure duration and seizure frequency logged by the seizure detection system can aid clinicians in deciding the treatment regime simultaneously. Therefore, it will be advantageous to detect seizures of short duration as well. In addition, a seizure detection system capable of detecting different types of convulsive seizures using short duration signal ($\geq$10 s) may help to understand patient specific seizure aetiology and might aid in reducing the burden on clinicians and epilepsy scientists. To elaborate the efficacy of the proposed approach, a thorough performance evaluation and comparison of the proposed approach with existing studies is presented here.

A. Performance of the Proposed Approach

For the patient group $PG1$ (with only GTCS events), one seizure event remained undetected (Table IV(a)) for patient P_{11}. Interestingly, P_{11} had multiple types of seizures (2 GTCS and 1 CPS) as seen from Table I. From visual inspections, it was found that the missed GTCS event was comparatively less violent and did not have the sudden bursts of activity typically associated with GTCS events [18], [36]. Therefore, the reason for missing this GTCS event can be attributed to the manifestation of this event, which was more tonic in nature and did not have the repetitive high frequency jerks typically associated to a GTCS event. In addition to P_{11}, patient P_5 also had multiple types of seizures (2 GTCS, and 1 PNES). Although all GTCS events were detected for P_5, it had shown the highest FAR (1.76/24 h) in comparison to other patients of the group $PG1$. Since patients with PNES suffers from an underlying psychoses [2], involuntary moves (which are not seizure events) may have contributed to this increased FAR for P_5. This indicates that seizure detection sensitivity and FAR both gets affected in case of patients with multiple seizure types.

A total of three seizure events were missed for patient group $PG2$, the three events corresponds to patients P_{11} (1 CPS), P_{13} (1 CPS), and P_{14} (1 CPS) as seen from Table IV(b). All the three missed events were CPS, which pronounces the similarity of convulsive CPS events to non-seizure events in comparison to GTCS events, and elucidates a possible reason of missing these seizures as normal/natural movements. In addition, only three patients with a total of 5 CPS events are present in our data set. Therefore, in LOPO validation process lesser number of CPS events are available for learning to build an optimum description, especially when the left-out patient is the one with CPS events.

For the patient group $PG3$, a total of five seizure events (all PNES) were undetected as seen from Table IV(c). The undetected events corresponded to P_{16}, P_{17} and P_{19}. For patient P_{17}, a total of 3 events were undetected, which were found to be less vigorous in their manifestation. In addition, patient P_{17} alone had 9 PNES events, which constitute approximately half of the total PNES events (9 out of 20 events (Table I)) used in this study. Therefore, during LOPO validation for patient P_{17} only half of the PNES events were available to train and find an optimal model. Thus, the selected model could not account for variability in PNES events. However, the same observations were not true for patients P_{16} and P_{19}. Further analysis of individual events of patients P_{16} and P_{19} in the feature space revealed that all the features (Table III) were non-discriminative, except the Poincaré descriptors. Use of only Poincaré features showed increased sensitivity (100%) for P_{16} and P_{19}. This improved detection sensitivity highlights the robustness of Poincaré based descriptors in classifying PNES events from non-seizure events. It would be interesting to investigate the efficacy of Poincaré based features on a larger group of patients with convulsive PNES.

A total of six events were missed by the algorithm for patient group $PG4$ as seen from Table IV(d). The missed events correspond to patients P_{11} (1 CPS), P_{16} (1 PNES), P_{17} (3 PNES), and P_{19} (1 PNES). Interestingly, both the patients as well the corresponding missing events are same, which was reported for patient group $PG1$ and $PG3$. Although, one CPS event is undetected for patient P_{11}, the other CPS events corresponding to patients P_{13} and P_{14} from group $PG2$ are detected here. These

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results indicate that use of PNES events for model training improves the detection sensitivity of CPS events, while remaining the same for GTCS and PNES events. However, this improved sensitivity also increases the FAR for both patients (see Table IV(b) and Table IV(d) for patients P_{13} and P_{14}). Another interesting observation that can be made from the validation results across the four groups (Table IV(a), IV(b), IV(c), and IV(d)) is that, all GTCS events are successfully detected when multiple seizure types in addition to GTCS are analyzed. Thus, a seizure detection system that is sensitive to multiple type of convulsive seizures is likely to have a better GTCS detection sensitivity, in comparison to systems developed using GTCS events only. Although these findings are interesting, no firm conclusion can be made at this stage with respect to CPS patients, since we only had 5 CPS events from 3 patients. Therefore, further investigation with a larger group of CPS patients is required to make a reliable conclusion with respect to CPS events. However, the inclusion of CPS patients in the study highlights an important group of patients having multiple epileptic disorders.

The trade-off between sensitivity and FAR across the four patient groups, can be explained using the SVDD description (Fig. 2). The radius of the hyper spherical decision boundary decreases and the number of support vectors (nSV) increase as we add CPS or/and PNES events with GTCS events (Fig. 2(a)–Fig. 2(d)). The reason behind increased number of support vector can be attributed to the fact that PNES and CPS events in comparison to GTCS are relatively less separable from non-seizure events. However, the effect of adding PNES events (Fig. 2(c)) on the decision boundary is much higher than CPS events (Fig. 2(b)). This indicates that, PNES events are more close to non-seizure events compared to CPS events in the SVDD feature space. However, further validation using higher number of CPS events is necessary to support this finding. The reason behind increased FAR can be attributed to the higher weight on sensitivity in the cost function (13), which leads to the reduction in volume of the spherical region enclosing the non-seizure events and causes an increase in FAR of the system. Therefore, the level of sensitivity and FAR can be controlled by the weight factor used in the cost function (13). However, the choice of higher weights for sensitivity is intentional as we do not want to miss any convulsive seizure.

The proposed algorithm detected 43 (84.31%) of 51 convulsive seizures with a total of 309 false alarms (1.33 per 24 h) when evaluated on 5,563 h of recording from right hand data of 79 patients (see details in supplementary material). In contrast to the right hand, a marginally higher sensitivity and a lower FAR (sens: 86.95%, and 270 false alarms (1.16 per 24 h)) is achieved for the left hand data. The results indicate that the seizure detection sensitivity of the proposed algorithm is largely unaffected by the choice of the arm while, the FAR can vary across arms. The higher FAR of the right hand data may be attributed to the fact, that most people use right hand for performing their daily activities [37]. Overall, these results indicate that seizures can be detected using a single wrist-worn accelerometer based device.

B. Comparison With Existing Seizure Detection Systems

Existing studies focus mainly on detection of GTCS thus, to elaborate the efficacy of the proposed approach the results for patient group $PG1$ are used for carrying out comparisons. Therefore, if we fix the type of seizure to be detected, then the three main contributors to the performance of a seizure detection system will be: 1) the minimum seizure duration that can be detected, 2) the number of sensors, and 3) multi-modality approaches. Therefore, it is only justifiable to have a comparison of the proposed approach by fixing one contributor across the studies and varying others. The performance is compared in terms of seizure detection sensitivity and FAR with publications on GTCS detection that use a non-EEG based wearable sensor (accelerometer, sEMG, EDA) system, and long-term continuous monitoring real-life data.

The sensitivity and FAR of a seizure detection system is dependent on the minimum seizure duration that the system can detect. Therefore, a comparison of seizure detection systems with a similar seizure duration will be warrant to the performance of the proposed system. In this work, we have used a minimum seizure duration of $\geq$ 10 s and only Milošević *et al.* [10], and Poh *et al.* [9] have presented an approach based on similar seizure duration (Table V). Both, Milošević *et al.* and Poh *et al.* investigated the use of multiple sensors in detection of GTCS events. Milošević *et al.* achieved an FAR of 1.94/24 h and 1.82/24 h with a detection sensitivity of 86% and 72.73% using two and four accelerometer sensors, respectively. Whereas, the best performance was achieved using two sEMG senors (FAR 0.56/24 h, Sens 82.00%). We could achieve a comparable performance (FAR 0.64/24 h, Sens 95.23%), using only a single wrist-worn accelerometer sensor. Milošević *et al.* [10] monitored pediatric patients who generally gets involved in more activities, which may contribute towards increased FAR. In addition, their approach is based on SVM, which might not be a suitable learning technique for a data set affected by class imbalance [27]. In contrast, a high seizure detection sensitivity is achieved using the proposed approach, where a learning technique (SVDD) that can handle the class imbalance in the data is employed [28]. Further, the proposed approach also showed a comparable performance to the study by Poh *et al.* [9]. They reported an FAR of 0.74/24 h and a seizure detection sensitivity of 94%. They employed an EDA sensor along with an accelerometer sensor; however, we achieved a similar seizure detection sensitivity with a lower FAR (FAR 0.64/24 h, Sens 95.23%) using a single accelerometer sensor. Although, Poh *et al.* [9], and Milošević *et al.* [10] have used multiple sensors, we could achieve a comparable performance using only a single sensor. This shows that the use of multiple sensors may not necessarily lead to an increase in the performance of the detection system.

Table VI compares the performance of our approach with other seizure detection systems, when number of sensors is fixed to a single sensor. The proposed system showed a comparable overall performance (sensitivity and FAR) to seizure detection systems based on a single accelerometer sensor (Beniczky *et al.* [15], Lockman *et al.* [16], and Kramer *et al.* [38]). First, our approach missed only one GTCS

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TABLE V
THE COMPARISONS OF THE PROPOSED STUDY, WITH STUDIES HAVING A SEIZURE DURATION OF ≥ 10 S

Study	*Proposed approach*	Milošević *et al.* [10]			Poh *et al.* [9]
Fixed parameters					
# seizure duration	$\geq 10s$	$\geq 10s$	$\geq 10s$	$\geq 10s$	$\geq 10s$
Variable parameters					
# No. of sensors	1 ACM	2 ACM	2 sEMG	4 ACM	1ACM+EDA
Total patients	79	56	56	56	80
Seizure patients	12	7	7	7	7
Total seizures	21	22	22	22	16
Seizure type	GTCS	GTCS	GTCS	GTCS	GTCS
Record duration	5576*h*	1998*h*	1998*h*	1998*h*	4213*h*
Sensitivity	95.23% (20/21)	86% (19/22)	82.00% (18/22)	72.73% (16/22)	94.00% (15/16)
FAR[¶]	0.64/24*h*	1.94/24*h*	0.56/24*h*	1.82/24*h*	0.74/24*h*

[¶] We have reported the FAR for all 79 patients corresponding to a record duration of 5576*h*.

TABLE VI
THE COMPARISONS OF THE PROPOSED STUDY, WITH SEIZURE DETECTION SYSTEMS BASED ON A SINGLE SENSOR

Study	*Proposed approach*	Beniczky *et al.* [15]	Lockman *et al.* [16]	Kramer *et al.* [38]	Conradsen *et al.* [11]	Szabó *et al.* [12]
Fixed parameters						
# No. of sensors	1 ACM	1 ACM	1 ACM	1ACM	1 sEMG	1 sEMG
Variable parameters						
# seizure duration	$\geq 10s$	−−	−−	$\geq 20s$	$\geq 50s$	−−
Total patients	79	73	40	31	60	33
Seizure patients	12	20	6	15	11	11
Total seizures	21	39	8	22	22	21
Seizure type	GTCS	GTCS	GTCS	C, GTCS	GTCS	GTCS
Record duration	5576*h*	4878*h*	−−	1692*h*	776*h*	1399*h*
Sensitivity	95.23% (20/21)	90.00% (35/39)	88.00% (7/8)	91% (20/22)	100% (22/22)	95% (20/21)
FAR[†]	0.64/24*h*	0.2/24*h*	−−	0.11/24*h*	0.96/24*h*	0.02/24*h*

[†] In the reference studies, FAR is calculated for total number of patients. Therefore, we have reported the FAR for all 79 patients corresponding to a record duration of 5576*h*.

event (mean sensitivity 95.83%; 95% [CI] 87.5–100%); however, the overall FAR (0.64/24 h) of the proposed approach is relatively higher in comparison to [15] and [38]. Second, the mean sensitivity of the proposed approach is 95.83% (95% [CI] 87.5–100%) compared to 90% and 91% as reported by [15] and [38]. However, the study by Beniczky *et al.* [15] is a prospective study. Third, in comparison to the system proposed by Kramer *et al.* [38], we used a smaller seizure duration which can be a reason for the increased FAR in our approach. As seizure duration is not mentioned in the study by Beniczky *et al.* [15], it is difficult to point out the exact reason for the lower FAR value in [15]. In contrast, the sEMG based system as reported by Conradsen *et al.* [11] detected all the seizures, and showed a higher detection sensitivity in comparison to the proposed approach. The improved seizure detection sensitivity can be attributed to the increased seizure duration of ≥ 50 s. However, the FAR of the sEMG based system [11] (0.96/24 h) is higher than the proposed approach (0.64/24 h). Although sEMG signal has a high specificity the reasons for a higher FAR (0.96/24 h), can be ascribed to the use of manual thresholding methods for detection of seizure events. Considering a lower FAR and only one missed seizure event, it can be argued that the performance of the proposed approach is comparable to the system proposed by Conradsen *et al.* [11]. On the other hand in comparison to [12], a similar seizure detection sensitivity of 95.23% (20/21) could be achieved by the proposed system. However, Szabó *et al.* [12] reported only 1 false alarm during 1, 399 h of monitoring period while, we obtained 149 false alarms in 5, 576 h of monitoring. The lower number of false alarms in the study by Szabó *et al.* [12] could be the result of conducting the study in controlled environment. The proposed system shows comparable performance to the best performing seizure detection systems based on a single sensor, which shows the potential of the proposed system to be used in ambulatory setting.

Several authors have investigated the use of multi-modality system for detection of seizures [9], [10]. Table VII shows the comparison of the proposed approach to multi-modality systems. It can be seen from Table VII, that the proposed systems is able to match the performance of multi-modality systems. Only 1 GTCS event (sensitivity (95.23%)) goes undetected using the proposed seizure detection system while, 1 (sensitivity (94%)) and 2 (sensitivity (91%)) events, respectively were missed using multi-modality approaches by [9], [10]. Therefore, the performance of the proposed approach in terms of the number of missed events, is comparable to the studies [9], [10]. In addition, the proposed approach has the lowest FAR (0.64/24 h) in comparison to the FAR of (0.74/24 h), and (1.0/24 h) reported by the multi-modality approaches [9], [10]. The performance of the proposed approach can be attributed to the (1) Poincaré based approach's, capturing nonlinear features and seizure event

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TABLE VII
THE COMPARISONS OF THE PROPOSED STUDY BASED ON SINGLE ACCELEROMETER SENSOR, TO MULTI-MODALITY SEIZURE DETECTION SYSTEMS

Study	*Proposed approach*	Poh *et al.* [9]	Milošević *et al.* [10]
Fixed parameters			
# seizure type	GTCS	GTCS	GTCS
Variable parameters			
# No. of sensors	1ACM	1ACM+EDA	2ACM+2sEMG
# seizure duration	$\geq 10s$	$\geq 10s$	$\geq 10s$
Total patients	79	80	56
Seizure patients	12	7	7
Total seizures	21	16	22
Record duration	5576*h*	4213*h*	1998*h*
Sensitivity	95.23% (20/21)	94.00% (15/16)	91.00% (20/22)
FAR†	0.64/24*h*	0.74/24*h*	1.0/24*h*

† In the reference studies, FAR is calculated for total number of patients. Therefore, we have reported the FAR for all 79 patients corresponding to a record duration of 5576*h*.

TABLE VIII
TABLE SHOWS THE BASELINE CHARACTERISTICS OF THE PATIENT COHORT, THE NUMBER OF OBSERVED SEIZURES, AS WELL AS THE NUMBER OF DETECTED SEIZURES AND THE NUMBER OF FALSE ALARMS PER PATIENT OBTAINED USING THE PROPOSED ACCELEROMETER BASED SYSTEM

Patient	Types of Seizure	Observed seizures (VEM)	Detected Seizures (ACM)	False alarms
P_1	1 GTCS	1	1	4
P_2	1 GTCS	1	1	7
P_3	1 GTCS	1	1	5
P_4	6 GTCS	6	6	3
P_5	2 GTCS, 1 PNES	3	3	10
P_6	2 GTCS	2	2	4
P_7	2 GTCS	2	2	3
P_8	1 GTCS	1	1	6
P_9	1 GTCS	1	1	2
P_{10}	1 GTCS	1	1	7
P_{11}	2 GTCS, 1 CPS	3	2	8
P_{12}	1 GTCS	1	1	3
P_{13}	3 CPS	3	3	5
P_{14}	1 CPS	1	1	9
P_{15}	5 PNES	5	5	6
P_{16}	1 PNES	1	0	4
P_{17}	9 PNES	9	6	3
P_{18}	1 PNES	1	1	3
P_{19}	1 PNES	1	0	6
P_{20}	2 PNES	2	2	18

where, GTCS: Generalized tonic-clonic seizures, PNES: Convulsive psychogenic non-epileptic seizures, CPS: Complex partial epileptic seizures, VEM: video electroencephalography monitoring, and ACM: accelerometer.

dynamics, and (2) the inclusion of SVDD classification. Moreover, the proposed system overcomes the inherent limitations of the multi-modality systems, like: non-obtrusive recording (no requirement of direct current to take measurements as in case of EDA), electrode-less system (less encumbering and more comfortable during continuous monitoring), and a simpler system (no requirement for complex integration of data from different sensors). However, the use of multi-modality systems especially the systems incorporating sEMG, may improve the detection of other seizure types especially, PNES [39].

Although, wireless remote monitoring devices have shown good clinical utility in management of epilepsy they are often reported to suffer from higher FAR [4]. Due to the inherent trade-off between seizure detection sensitivity and FAR, it is often difficult to avoid false positives. In this study, we achieved a mean sensitivity of 86.95% with a total of 116 false alarms from 20 patients having convulsive seizures (GTCS, PNES, and CPS). We reviewed the VEM recording of patients with seizures to identify if any particular task or activity is triggering the false alarms. We could only review video data corresponding to 8 out of 20 patients because of policy on epilepsy monitoring unit to cut down the non-seizure parts of the video before storing. In addition, we did not review the recorded false positive events when the patient is not in the camera's field of view and/or when the video resolution is poor, such that, the activity could not be ascertained. Table VIII shows the observed seizures (VEM), as well as the number of false alarms and the number of seizures detected using the proposed system. In terms of false alarms, patients P_{20} and P_{11} (out of 8 reviewed patients) had the majority of false alarms, contributing to approximately one-fourth of total false alarms (Table VIII). From the VEM, it was observed that patient P_{20} was involved in continuous excessive movements that lead to false alarms: shifting positions, swinging and crossing arms, fiddling with the strap of the device, taking off the device while lying down, and sleeping on top of the device. However, for patient P_{11} the events that triggered false alarms were: repetitive taping of left upper and lower limb, scribbling and writing on a notepad, playing games on notepad, and fumbling with a strap containing electrodes. On the other hand, for rest of patients (P_1, P_2, P_3, P_8, P_{15}, and P_{17}) the events that triggered false alarms were mostly due to non-purposeful movements, such as: fidgeting, making gestures while talking (supination and pronation of wrist), continuously changing postures while lying down, stretching and crossing arms, rubbing hands, swiping on mobile screen, and an event where the patient is turning and flipping the pages of a book. From the above analysis, we noticed that few false alarms (non-purposeful gestures, changing postures while lying down) were triggered in clusters while, none showed high correlation to any specific ADL. However, activities associated to restlessness (continuously changing postures, swinging and crossing arms) and some habitual movements (taping of limbs, gestures while talking) that triggered false alarms, were more pronounced across the patients. This indicates that (1) the performance of the proposed system is not influenced or affected by any particular ADL, and (2) the resulting false positives were majorly attributed to out of context or non-purposeful activities. In future, these findings can be incorporated to design better filtering strategies to minimize FAR of the seizure detection system.

The proposed study is focused on detection of different types of convulsive seizures. In addition to detection, classification of different convulsive seizures will further enhance the utility of the proposed system and aid in reducing the burden of the expensive differential diagnosis procedures. In future work, we will investigate approaches to classify different convulsive epileptic seizures. However, the proposed seizure detection system based on accelerometry can have the following limitations:

- The system is developed and tested on data collected in a hospital setting. The system is still to be validated in patients natural setting.
- The system, is responsive to convulsive seizures that manifest in arms. Results might vary with the position of device.
- Non-convulsive seizures will go undetected.

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- A small group of 3 patients is used for CPS analysis. Further, analysis on a larger group of patients with CPS is required to strengthen the current findings.

V. Conclusion

We proposed a wrist-worn wireless seizure detection system for remote monitoring of epileptic patients. The proposed system uses a single accelerometer sensor, and is capable of detecting convulsive seizures with duration $\geq$ 10 s. The proposed algorithm is validated on accelerometer data collected from patients undergoing VEM. A new set of computationally efficient time domain features, including features extracted using a nonlinear method (Poincaré plot) were used for training the SVDD classifier. The algorithm detected 40 (86.95%) of 46 convulsive seizures with a mean FAR of 2.25/24 h. The comparison with existing GTCS detection systems shows the feasibility of the proposed approach (GTCS detection sensitivity (mean 95.83%; 95% [CI] 87.5–100%) and FAR 0.64/24 h) in automated detection of seizures. Furthermore, results from the individual analysis of the left and right hand data indicate the feasibility of a single wrist-worn accelerometer device for the detection of convulsive seizures. The results demonstrate the system's potential in detecting seizures in both clinical and non-clinical environments.

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References

[1] World Health Organization, "Epilepsy: Fact sheet," Feb. 2017. http://www.who.int/mediacentre/factsheets/fs999/en/
[2] J. Bayly *et al.*, "Time-frequency mapping of the rhythmic limb movements distinguishes convulsive epileptic from psychogenic nonepileptic seizures," *Epilepsia*, vol. 54, no. 8, pp. 1402–1408, 2013.
[3] R. Surges *et al.*, "Sudden unexpected death in epilepsy: Risk factors and potential pathomechanisms," *Nature Rev. Neurol.*, vol. 5, no. 9, pp. 492–504, 2009.
[4] C. Jory *et al.*, "Safe and sound? A systematic literature review of seizure detection methods for personal use," *Seizure*, vol. 36, pp. 4–15, 2016.
[5] A. Schulze-Bonhage *et al.*, "Views of patients with epilepsy on seizure prediction devices," *Epilepsy Behavior*, vol. 18, no. 4, pp. 388–396, 2010.
[6] K. Jansen *et al.*, "Peri-ictal ECG changes in childhood epilepsy: Implications for detection systems," *Epilepsy Behavior*, vol. 29, no. 1, pp. 72–76, 2013.
[7] S. Ramgopal *et al.*, "Seizure detection, seizure prediction, and closed-loop warning systems in epilepsy," *Epilepsy Behavior*, vol. 37, pp. 291–307, 2014.
[8] M.-Z. P , "Autonomic changes with seizures correlate with postictal EEG suppression," *Neurology*, vol. 78, no. 23, pp. 1868–1876, 2012.
[9] M.-Z. Poh *et al.*, "Convulsive seizure detection using a wrist-worn electrodermal activity and accelerometry biosensor," *Epilepsia*, vol. 53, no. 5, pp. e93–e97, 2012.
[10] M. Milošević *et al.*, "Automated detection of tonic clonic seizures using 3-D accelerometry and surface electromyography in pediatric patients," *IEEE J. Biomed. Health Inf.*, vol. 20, no. 5, pp. 1333–1341, Sep. 2016.
[11] I. Conradsen *et al.*, "Automated algorithm for generalized tonic clonic epileptic seizure onset detection based on SEMG zero-crossing rate," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 2, pp. 579–585, Feb. 2012.
[12] C. Á. Szabó *et al.*, "Electromyography-based seizure detector: Preliminary results comparing a generalized tonic–clonic seizure detection algorithm to video-eeg recordings," *Epilepsia*, vol. 56, no. 9, pp. 1432–1437, 2015.
[13] I. Conradsen *et al.*, "Evaluation of novel algorithm embedded in a wearable SEMG device for seizure detection," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.*, 2012, pp. 2048–2051.
[14] K. Cuppens *et al.*, "Accelerometry-based home monitoring for detection of nocturnal hypermotor seizures based on novelty detection," *IEEE J. Biomed. Health Inf.*, vol. 18, no. 3, pp. 1026–1033, May 2014.
[15] S. Beniczky *et al.*, "Detection of generalized tonic–clonic seizures by a wireless wrist accelerometer: A prospective, multicenter study," *Epilepsia*, vol. 54, no. 4, pp. e58–e61, 2013.
[16] J. Lockman *et al.*, "Detection of seizure-like movements using a wrist accelerometer," *EpilepsyBehavior*, vol. 20, no. 4, pp. 638–641, 2011.
[17] G. Becq *et al.*, "Classification of epileptic motor manifestations and detection of tonic–clonic seizures with acceleration norm entropy," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 8, pp. 2080–2088, Aug. 2013.
[18] E. Schulc *et al.*, "Measurement and quantification of generalized tonic-clonic seizures in epilepsy patients by means of accelerometry? an explorative study," *Epilepsy Res.*, vol. 95, no. 1, pp. 173–183, 2011.
[19] J. van Andel *et al.*, "Non-eeg based ambulatory seizure detection designed for home use: What is available and how will it influence epilepsy care?" *Epilepsy Behavior*, vol. 57, pp. 82–89, 2016.
[20] M. Reuber *et al.*, "Failure to recognize psychogenic nonepileptic seizures may cause death," *Neurology*, vol. 62, no. 5, pp. 834–835, 2004.
[21] S. G. Jones *et al.*, "Clinical characteristics and outcome in patients with psychogenic nonepileptic seizures," *Psychosomatic Med.*, vol. 72, no. 5, pp. 487–497, 2010.
[22] P. Kotagal *et al.*, "Complex partial seizures of frontal lobe onset statistical analysis of ictal semiology," *Seizure*, vol. 12, no. 5, pp. 268–281, 2003.
[23] T. M. Nijsen *et al.*, "Detection of subtle nocturnal motor activity from 3-D accelerometry recordings in epilepsy patients," *IEEE Trans. Biomed. Eng.*, vol. 54, no. 11, pp. 2073–2081, Nov. 2007.
[24] A. Dalton *et al.*, "Development of a body sensor network to detect motor patterns of epileptic seizures," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 11, pp. 3204–3211, Nov. 2012.
[25] T. M. E. Nijsen *et al.*, "Time-frequency analysis of accelerometry data for detection of myoclonic seizures," *IEEE Trans. Inf. Technol. Biomed.*, vol. 14, no. 5, pp. 1197–1203, Sep. 2010.
[26] I. Conradsen *et al.*, "Automatic multi-modal intelligent seizure acquisition (MISA) system for detection of motor seizures from electromyographic data and motion data," *Comput. Methods Programs Biomed.*, vol. 107, no. 2, pp. 97–110, 2012.
[27] R. Batuwita and V. Palade, "Class imbalance learning methods for support vector machines," *Imbalanced Learning: Foundations, Algorithms, and Applications*, vol. 83. Hoboken, NJ, USA: Wiley, 2013.
[28] D. M. Tax and R. P. Duin, "Support vector data description," *Mach. Learn.*, vol. 54, no. 1, pp. 45–66, 2004.
[29] J. Gubbi *et al.*, "Automatic detection and classification of convulsive psychogenic non-epileptic seizures using a wearable device," *IEEE J. Biomed. Health Informat.*, vol. 20, no. 4, pp. 1061–1072, Jul. 2016.
[30] M. Milošević *et al.*, "Feature selection methods for accelerometry-based seizure detection in children," *Med. Biol. Eng. Comput.*, vol. 55, pp. 151–165, 2016.
[31] L. Logesparan *et al.*, "Optimal features for online seizure detection," *Med. Biol. Eng. Comput.*, vol. 50, no. 7, pp. 659–669, 2012.
[32] A. H. Khandoker *et al.*, *Poincaré Plot Methods for Heart Rate Variability Analysis*. Berlin, Germany: Springer, 2013.
[33] C. K. Karmakar *et al.*, "Complex correlation measure: A novel descriptor for poincaré plot," *Biomed. Eng. Online*, vol. 8, no. 1, p. 17, 2009.
[34] J. A. Hanley and B. J. McNeil, "A method of comparing the areas under receiver operating characteristic curves derived from the same cases." *Radiology*, vol. 148, no. 3, pp. 839–843, 1983.
[35] D. M. Tax and R. P. Duin, "Support vector domain description," *Pattern Recognit. Lett.*, vol. 20, no. 11, pp. 1191–1199, 1999.
[36] T. M. Nijsen *et al.*, "The potential value of three-dimensional accelerometry for detection of motor seizures in severe epilepsy," *Epilepsy Behavior*, vol. 7, no. 1, pp. 74–84, 2005.
[37] R. C. Oldfield, "The assessment and analysis of handedness: The edinburgh inventory," *Neuropsychologia*, vol. 9, no. 1, pp. 97–113, 1971.
[38] U. Kramer *et al.*, "A novel portable seizure detection alarm system: Preliminary results," *J. Clin. Neurophysiol.*, vol. 28, no. 1, pp. 36–38, 2011.
[39] S. Beniczky *et al.*, "Quantitative analysis of surface electromyography during epileptic and nonepileptic convulsive seizures," *Epilepsia*, vol. 55, pp. 1128–1134, 2014.

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